Lab 3: Evaluate Your Data Agent

Pre-requisite

1. You have completed Lab 1 & Lab 2.

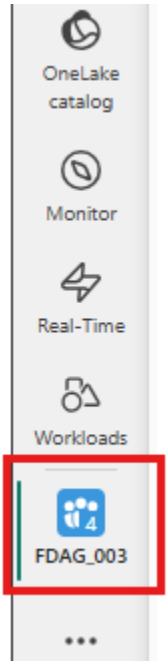
Learning Objectives

1. Apply your learning of Prompt engineering to build the best Data Agent possible for the Data Set
2. Use evaluation notebook to evaluate the quality of your Data Agent, and iteratively improve on your Data Agent using evaluation notebook results.

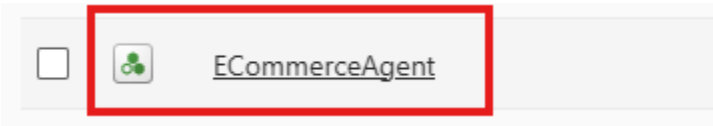
Lab Steps

Step 1: Create a NEW Data Agent

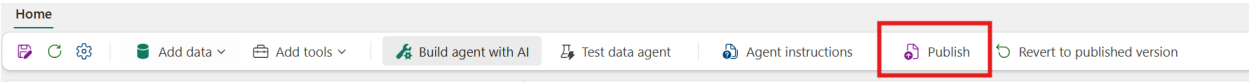
1. Return to your workspace through the Multi-tasking Nav:



2. Open your Data Agent:



3. Publish the Data Agent

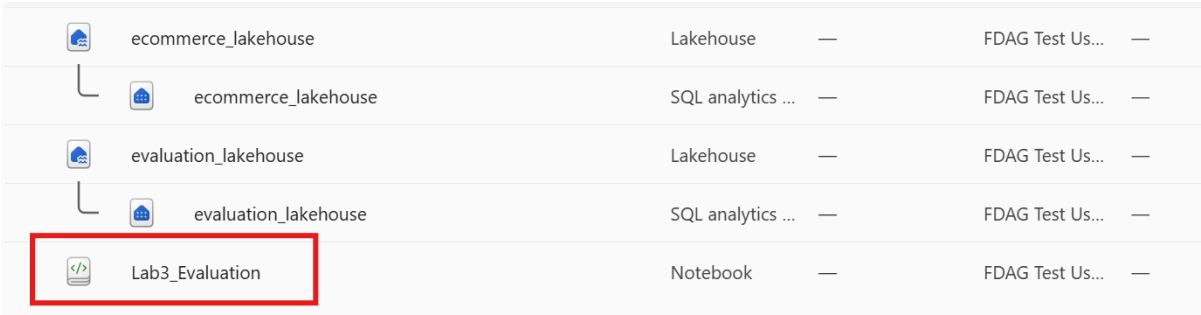


Publishing the Data Agent allows Data Agent to be consumed by downstream channels. In this lab, we will use the published data agent and run an evaluation across benchmark questions, and evaluate how well the Data Agent performs.

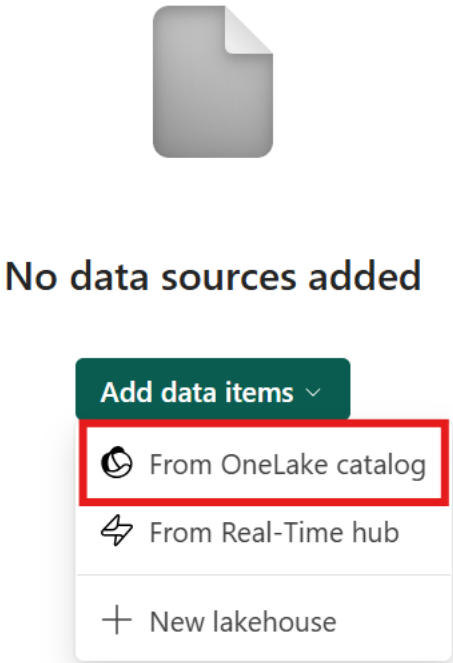
Note: This is a very important step. The Evaluation executes against a **published** Data Agent. This means every time you want to execute the evaluation with any changes to Data Agent's configuration, you must publish the Data Agent.

Step 2: Load the Evaluation notebook, and run it

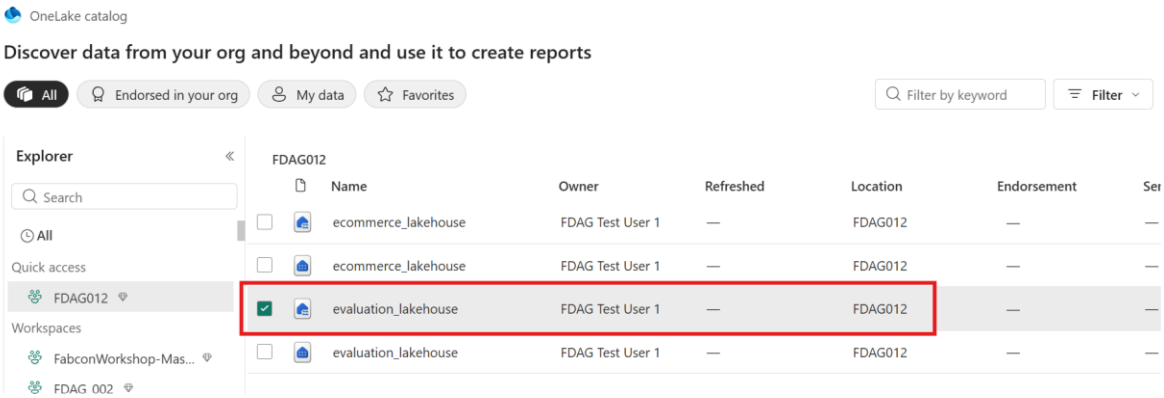
1. Open the Evaluation Notebook “Lab3_Evaluation”:



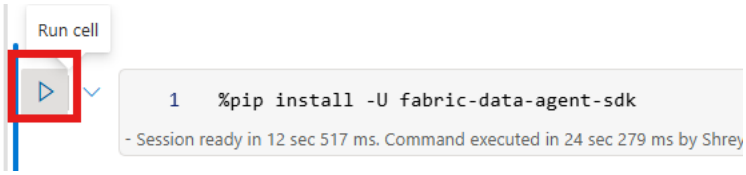
2. Select “Add data Items” -> “From OneLake catalog”



3. Select “evaluation_lakehouse”-> “Add”



4. Start this lab by clicking the “Run” Button in the Fisrt cell with “pip install” in it:



5. Update:
   “YOUR_DATA_AGENT_ARTIFACT_NAME” variable with your Data Agent’s artifact name.
   “YOUR_ACCOUNT_NUMBER” variable with your 3 digit account number

STEP 1: Use the Evaluation Notebook

1. **Update the `YOUR_DATA_AGENT_ARTIFACT_NAME`**
   - Replace the placeholder with the name of your Data Agent Artifact from in Step 1 of the "Lab 3 Curate Your Data Agent" Document.
2. **Update the `YOUR_ACCOUNT_NUMBER` Variable**
   - Replace the placeholder with your name

```
from fabric.dataagent.client import FabricDataAgentManagement
from fabric.dataagent.evaluation import evaluate_data_agent, get_evaluation_summary, get_evaluation_details, get_evaluation_summary_per_question
#from fabric.dataagent.evaluation.evaluator_api import add_ground_truth_batch
import openai
import os
import re

YOUR_DATA_AGENT_ARTIFACT_NAME = "EvalAgent-AGNT200"

YOUR_ACCOUNT_NUMBER = "200"

os.environ["OPENAI_API_VERSION"] = "2023-05-15"

data_agent = FabricDataAgentManagement(YOUR_DATA_AGENT_ARTIFACT_NAME)
```

✓ - Command executed in 18 sec 530 ms by Shreyas Canchi Radhakrishna on 10:55:47 PM, 9/08/25

   a. After this step, you should be able to run through each step in the notebook. Please follow along and read the directions as you go!
   b. You should have been able to run all the cells in the Notebook, and get a result from the evaluation.

Step 3: Improve Configuration Set Up and Pass All Evaluation

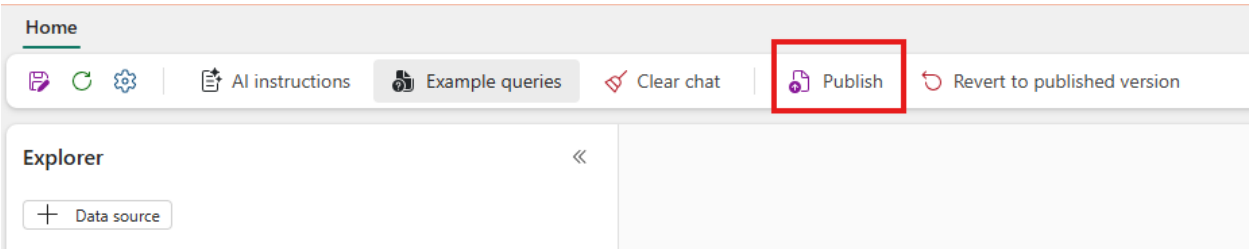
1. Identify all the Questions the Evaluation failed in:

index	T	F	?	%	failed_threads	question
0	0	1	0	0.0	1	Compute the median product price for category beauty and health.
1	0	0	1	0.0	1	Find the top city by the number of delivered orders.
2	0	0	1	0.0	1	Give me the revenue from furniture related purchases

Note: While the links above allow you to view a chat thread with the Data Agent, this link will only provide a **Consumer View**. This means that you will not be able to edit

ANY configurations. To make configuration changes, open your Data Agent from your Workspace.

2. Open your Data Agent and update the configurations in the Data Agent to address the failed questions. Use the tools you discovered in Lab 2:
    a. Add Agent Instructions
    b. Add Data Source Instructions
    c. Add Example Queries
3. Publish the Data Agent:



4. Re-run the "📊 STEP 3: Evaluation" python cell in the Evaluation Notebook:

```python

# The table used to store evaluation results
evaluation_table_name = f'evaluation_output_{YOUR_NAME}'

# run evaluation

eval_id = evaluate_data_agent(
    df = df_pandas_groundtruth,
    data_agent_name = YOUR_DATA_AGENT_ARTIFACT_NAME,
    data_agent_stage = "production",
    table_name = evaluation_table_name)

df = get_evaluation_summary_per_question(eval_id, verbose=True, table_name = evaluation_table_name)
```

Goal: The Data Agent should answer ALL questions in the Evaluation notebook.

Step 4: (Optional) Add a new Benchmark

1. If you finish this lab early, try to add a new benchmark question to your data agent, or expand on the existing benchmarks by asking for different product_categories or columns to make your data agent more robust.

Congratulations, you've completed this lab!